Ian Leung

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Education

University of Waterloo **Expected May 2029**
Bachelor of Computer Science, Finance, and Statistics (Co-op) GPA: 3.8/4.0

- President's Scholarship of Distinction ($5000), Undergraduate Achievement Award ($1500)
- Advanced Coursework: Functional Programming, Sequential Programming, Object-Oriented Programming, Interpreters

Experience

Roblox **June 2026 – Aug 2026**
Software Engineer Intern, Network Infrastructure San Mateo, CA

- Cut deployment time to 1 hour across 5000+ production devices in global datacenters with a GitHub Actions CI/CD pipeline equipped with staged expansion and health checks
- Saved 400+ engineering hours annually and identified issues on 20+ network devices by building Temporal Workflows with LLM analysis and Slack alerting for autonomous rollout of configuration changes
- Recovered 7% CPU utilization across devices by identifying and fixing a resource leak in an internal network service

Tells.dev **Apr 2026 – June 2026**
Co-Founder Waterloo, ON

- Scaled startup from 0 to 14 paying customers in its first month, amassing 50k+ minutes of session time and generating interest from angel investors and VC firms
- Built browser infrastructure enabling AI agents to run autonomous web sessions on production websites

Sunnybrook Research Institute **July 2025 – Aug 2025**
Software Research Intern, Process Automation Toronto, ON

- Cut testing time by 91% by automating regression tests with synthetic JSON event flows and recorded baselines
- Surfaced 5+ defects and edge cases in real usage patterns by building an event tracing, playback, and logging tool that records and simulates clinician workflows in Focused Ultrasound software

Projects

Waterloo Quant Club: Software Engineer | AWS, Grafana, Rust, WebSocket **Jan 2026 – Present**

- Served 1300+ users with portal.waterlooquantclub.com, a full-stack club recruitment and event portal hosted on AWS
- Launched a Grafana observability platform with 100+ metrics, including membership and infrastructure monitoring
- Built and deployed 2v2bluff, a real-time multiplayer card game for 4 concurrent players using Rust and WebSocket

OuterwoRLd: Reinforcement Learning and 2D Physics Game | C++, Python, OpenGL, PyTorch, TensorBoard

- Raised success rate on newly generated levels from 3% to 96% by implementing Proximal Policy Optimization (PPO) from scratch in PyTorch to train a reinforcement learning agent
- Increased success on obstacle-avoidance levels from 1% to 100% by diagnosing a credit-assignment failure (0.3 s of effective lookahead time) and fixing it with an 8-step action repeat
- Built a 2D spaceflight physics game in C++ and OpenGL, with procedural level generation and gravity-based obstacles

Jevis: Browser-Use Agent Harness (Hack the North 2026 Winner) | Python, Playwright, JEV, OpenAI

- Cut token cost by 250x compared to GPT-6 Astra by building an agentic browser-use harness on the JEV System One Model that automates browser tasks through a web interface
- Reduced Walmart query time from 5 minutes to 44 seconds using a DOM-based approach and orchestrator models
- Achieved a 15% cache hit rate by building a caching layer for repeated agent tasks using precomputed instruction paths

Skills

Languages	Python, C++, Java, C#, Rust, TypeScript, SQL, Bash
Technologies	Temporal, gRPC, REST, OpenGL, PyTorch, Pandas, NumPy, OpenCV, PostgreSQL, LLMs, React
Tools	Git, Docker, AWS, GitHub Actions, Grafana, TensorBoard, AI Agents

Competitions

Galois Math Contest Honor Roll · Canadian Computing Competition Distinction · ICDC International Finalist